

```
# Import Required Libraries
```

```
import pandas as pd
```

```
import matplotlib.pyplot as plt
```

```
import seaborn as sns
```

```
# Set Visualization Style
```

```
plt.style.use('fivethirtyeight')
```

```
# Read Dataset
```

```
df = pd.read_csv("dataset/loan_data.csv")
```

```
# Display First Five Records
```

```
print(df.head())
```

```
# Display Dataset Information
```

```
print(df.info())
```

```
# -----
```

```
# Multivariate Analysis using Swarm Plot
```

```
# -----
```

```
plt.figure(figsize=(10,6))
```

```
sns.swarmplot(
```

```
x='Property_Area',  
y='ApplicantIncome',  
hue='Loan_Status',  
data=df  
)  
  
plt.title("Applicant Income by Property Area and Loan Status")  
plt.xlabel("Property Area")  
plt.ylabel("Applicant Income")  
  
plt.legend(title="Loan Status")  
  
plt.show()
```